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The extended-$\left(\frac{G'}{G}\right)$-expansion method and new exact solutions for the conformable space-time fractional diffusive predator-prey system

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This paper explores the dynamics of the conformable space-time fractional diffusive predator-prey system using the extended-$\left(\frac{G'}{G}\right)$-expansion method. By incorporating negative exponents into the solution process, the study uncovers a broader set of exact solutions, providing deeper insights into the system's behavior. The analysis reveals how the solutions' singularities shift in response to variations in the fractional orders, with a movement towards the boundary as the fractional order decreases and towards the angle bisector of space-time as it approaches 1. Additionally, the paper identifies chaotic phenomena that emerge in the triangular solution when the fractional order is near 0.7, highlighting the sensitivity of the system to small changes in this range. Through graphical representations and solution analysis, the paper demonstrates how these fractional-order dynamics influence predator-prey interactions, offering valuable contributions to both mathematical biology and the study of fractional differential equations in complex systems.

Keywords Predator-prey model, Fractional-order differential equations, Extended-$\frac{G'}{G}$-expansion method, Exact solutions

Nonlinear partial differential equations play a crucial role in modeling and analyzing various complex phenomena in science and engineering. In recent years, the introduction of fractional derivatives has significantly expanded the ability of differential equations to describe intricate real-world processes, particularly in physics, biology, and ecology[1–5]. Among these applications, the predator-prey model[6–10] has attracted considerable attention due to its ability to simulate ecological interactions and population dynamics.

The conformable space-time fractional diffusive predator-prey system

$$\begin{cases} D_t^{\mu}\phi_1 = D_{xx}^{\lambda}\phi_1 - s\phi_1 + (1+s)\phi_1^2 - \phi_1\phi_2 - \phi_1^3, \\ D_t^{\mu}\phi_2 = D_{xx}^{\lambda}\phi_2 - b\phi_2 + r\phi_1\phi_2 - \delta\phi_2^3, \end{cases} \tag{1}$$

where the parameters s, b, r, δ are positive constants, provides a more refined approach to modeling predator-prey interactions by incorporating fractional derivatives, which better capture memory effects and anomalous diffusion behaviors. However, due to the complexity of fractional differential equations, obtaining precise analytical solutions remains a challenging task. Traditional methods often fail to provide a comprehensive understanding of the system's dynamics, necessitating the development of more advanced analytical techniques.

In[11], the study models Hepatitis B virus (HBV) transmission using a fractional-order framework with the Caputo-Fabrizio derivative to capture memory effects and complex dynamics. Their key findings show that reducing the fractional order lowers infection rates, while higher vaccination rates significantly curb HBV

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spread. Numerical simulations highlight critical parameters like contact rates and vaccine efficacy, offering insights for optimizing public health interventions to control HBV. Similar work we refer to[12–19].

The paper[20] employs a Liouville-Caputo fractional derivative to analyze a compartmental model for computer virus propagation, emphasizing network security and malware dynamics. By integrating fixed-point theorems and numerical methods, the authors establish solution existence, uniqueness, and Ulam-Hyers stability. While the focus is on cybersecurity, the fractional framework captures memory effects and non-local interactions, akin to space-time fractional diffusive models used in biological systems (e.g., disease spread). Though not explicitly using conformable derivatives-a simpler fractional operator with local properties-the work highlights how fractional calculus enhances modeling of complex, memory-dependent processes. For other related work, see[21–27].

The paper[28] (see also[29–31]) introduces a hybrid method combining Caputo derivatives with Lucas-Fibonacci polynomials and Strang splitting to solve two-term fractional Sobolev equations. Validated on engineering and physics problems, the approach excels in accuracy for irregular domains. While the applications are non-biological, the methodology's treatment of space-time fractional diffusive models is transferable to biological contexts, such as anomalous diffusion in tissues or pollutant transport. The conformable derivative, noted for its simplicity and adherence to classical derivative rules, is not utilized here; instead, the non-local Caputo operator aligns with traditional fractional modeling of systems requiring memory effects.

Both studies[20] and[28] leverage fractional calculus to address non-local dynamics, with methodologies adaptable to biological systems. Space-time fractional models are implicitly relevant to biological diffusion processes (e.g., drug delivery, epidemiology), though the papers focus on cybersecurity and engineering. The conformable derivative-a distinct, local fractional operator-remains unexplored here but could offer computational advantages for real-time biological simulations.

The study[32] models HIV infection under combination antiretroviral therapy (cART) using a fractional framework to capture complex virus-immune interactions. Simulations show how treatment parameters influence viral behavior, highlighting the model's utility in optimizing cART strategies for improved outcomes. We refer to[32–37] for related work.

To address this challenge, this paper applies the extended-$\left(\frac{G'}{G}\right)$-expansion method to derive exact solutions for the conformable space-time fractional diffusive predator-prey system. Compared to previous studies, this approach allows for the introduction of negative exponents, leading to a broader class of exact solutions. These solutions enable a deeper analysis of the system's behavior and provide new insights into the evolution of predator-prey interactions under fractional-order dynamics.

To our best knowledge, there are no precise well-posedness results to provide for them. In[7], the $\left(\frac{G'}{G}\right)$-expansion was studied. For more relevant literature on fractional differential equations and , we refer to[38–44]. About the details of numerical solution method, we can refer to[45]. Since the extended-$\left(\frac{G'}{G}\right)$-expansion method[42,46] can obtain more exact solutions, we will focus on it in this paper. By using this method, we can obtain a more diverse class of traveling wave solutions, and then analyze the corresponding solutions to understand their evolutionary patterns.

By analyzing the obtained solutions through graphical representations, we examine how different fractional orders influence the regularity and singularity of solutions. Our findings reveal that as the fractional order decreases, the singularity of the solutions shifts toward the boundary of the region, while increasing the fractional order causes the singularity to move toward the angle bisector of space-time. Additionally, an intriguing chaotic behavior emerges when the fractional order is around 0.7, highlighting the system's sensitivity to small parameter changes. This study contributes to both the mathematical theory of fractional differential equations and the broader understanding of predator-prey dynamics. The results presented here provide valuable insights for researchers in mathematical biology, applied mathematics, and related fields. The remainder of the paper is structured as follows: Section 2 presents the application of the extended-$\left(\frac{G'}{G}\right)$-expansion method and the derivation of exact solutions. Section 3 discusses the graphical representations of the solutions and their dynamic behaviors. Finally, Section 4 provides the conclusion and key takeaways from this study.

Last, we explain the necessary notations and required terms used in this paper. The derivative D_{λ}^{μ} is the conformable space-time fractional derivative, see Definition 1. λ and μ $(0 < \lambda, \mu \leq 1)$ denote the space and time-fractional order derivative. $\phi_1 = \phi_1(x,t)$ and $\phi_2 = \phi_2(x,t)$ represent the prey and predator densities, respectively. To conveniently investigate the dynamics of the above diffusive predator-prey system (1), let $b = s, r + \frac{1}{\delta} = s + 1$, we can get the following form

$$\begin{cases} D_t^{\mu}\phi_1 = D_{xx}^{\lambda}\phi_1 - s\phi_1 + \left(r + \frac{1}{\delta}\right)\phi_1^2 - \phi_1\phi_2 - \phi_1^3, \\ D_t^{\mu}\phi_2 = D_{xx}^{\lambda}\phi_2 - s\phi_2 + r\phi_1\phi_2 - \delta\phi_2^3. \end{cases} \tag{2}$$

Exact solutions

In this section, we apply the extended-$\left(\frac{G'}{G}\right)$-expansion method to find the exact solutions of (1).

Conformable fractional derivatives

First, we give the definition of conformable fractional derivatives in[47]:

Definition 1 For a function $f:(0,+\infty)\longrightarrow\mathbb{R}$, the conformable fractional derivative of order μ is defined by

$$D_t^{\mu}f(t)=\lim_{\epsilon\to 0}\frac{f(t+\epsilon t^{1-\mu})-f(t)}{\epsilon}.$$

Remark 1 (Derivative rules) For any $\mu\in(0,1]$, we have

(1) $D_t^{\mu}(af(t)+bg(t))=aD_t^{\mu}f(t)+bD_t^{\mu}g(t)$;
(2) $D_t^{\mu}(t^q)=qt^{q-\mu}$ for all $q\in\mathbb{R}$;
(3) $D_t^{\mu}(f(t)g(t))=f(t)D_t^{\mu}g(t)+g(t)D_t^{\mu}f(t)$;
(4) $D_t^{\mu}\left(\frac{f(t)}{g(t)}\right)=\frac{g(t)D_t^{\mu}f(t)-f(t)D_t^{\mu}g(t)}{g^2(t)}$.

The Extended-$\left(\frac{G'}{G}\right)$-Expansion Method

To transfer the original system into the traditional form, we first use the traveling wave transformation (3) and Remark 1

$$\phi_1(x,t)=\phi_1(\xi),\quad \phi_2(x,t)=\phi_2(\xi),\quad \xi=\frac{x^{\lambda}}{\lambda}-c\frac{t^{\mu}}{\mu}. \tag{3}$$

to get the following ordinary differential equations (ODEs)

$$\begin{cases}\phi_1''+c\phi_1'-s\phi_1+\left(r+\frac{1}{\sqrt{\delta}}\right)\phi_1^2-\phi_1\phi_2-\phi_1^3=0,\\ \phi_2''+c\phi_2'-s\phi_2+r\phi_1\phi_2-\delta\phi_2^3=0.\end{cases} \tag{4}$$

In order to proceed, we assume the following analytical solutions, including the negative exponents:

$$\begin{cases}\phi_1(\xi)=\sum_{i=0}^{m}a_i\left(\frac{G'}{G}\right)^i+\sum_{i=-m}^{-1}a_i\left(\frac{G'}{G}\right)^i,\\ \phi_2(\xi)=\sum_{i=0}^{n}b_i\left(\frac{G'}{G}\right)^i+\sum_{i=-n}^{-1}b_i\left(\frac{G'}{G}\right)^i,\end{cases} \tag{5}$$

where G satisfies the assumed ODE:

$$\frac{G''}{G}=\alpha\left(\frac{G'}{G}\right)^2+\beta\frac{G'}{G}+\gamma. \tag{6}$$

When $\alpha=1$, its solution can be expressed as

$$\frac{G'(\xi)}{G(\xi)}=Ce^{\beta\xi}-\gamma\quad\text{for all } C\in\mathbb{R}. \tag{7}$$

Otherwise, letting $\Delta=\beta^2-4\alpha(\alpha-1)\gamma$, we obtain

$$\frac{G'(\xi)}{G(\xi)}=\begin{cases}\frac{\sqrt{\Delta}}{2(1-\alpha)}\left(\frac{C_1\sinh\left(\frac{\sqrt{\Delta}}{2}\xi\right)+C_2\cosh\frac{\sqrt{\Delta}}{2}\xi}{C_1\cosh\left(\frac{\sqrt{\Delta}}{2}\xi\right)+C_2\sinh\frac{\sqrt{\Delta}}{2}\xi}\right)+\frac{\beta}{2(1-\alpha)}, & \Delta>0,\ \alpha\neq 1;\\ \frac{\sqrt{-\Delta}}{2(1-\alpha)}\left(\frac{-C_1\sin\left(\frac{\sqrt{-\Delta}}{2}\xi\right)+C_2\cos\left(\frac{\sqrt{-\Delta}}{2}\xi\right)}{C_1\cos\left(\frac{\sqrt{-\Delta}}{2}\xi\right)+C_2\sin\left(\frac{\sqrt{-\Delta}}{2}\xi\right)}+\frac{\beta}{2(1-\alpha)}\right), & \Delta<0,\ \alpha\neq 1;\\ \frac{1}{1-\alpha}\left(\frac{C_1}{C_1\xi+C_2}+\frac{\beta}{2}\right), & \Delta=0,\ \alpha\neq 1.\end{cases} \tag{8}$$

The principle of homogeneous balance method[48] is now utilized to simplify (5). Indeed, we must have

$$\deg(\phi_i'')=\deg(\phi_i^3)\quad\text{for all } i=1,2. \tag{9}$$

Keep in mind the equation (6), one has

$$(\alpha-1)^2m(m+1)a_m\left(\frac{G'}{G}\right)^{m+2}=a_m^3\left(\frac{G'}{G}\right)^{3m}. \tag{10}$$

Equating the coefficients and the exponents of $\left(\frac{G'}{G}\right)$ from (10), we obtain

$$m=1\quad\text{and}\quad 2(\alpha-1)^2a_1=a_1^3. \tag{11}$$

Similarly, we can also get

$$n = 1 \quad \text{and} \quad 2(\alpha-1)^2 b_1 = b_1^3. \tag{12}$$

Consequently, (5) becomes

$$\begin{cases} \phi_1(\xi) = a_0 + a_1\left(\frac{G'}{G}\right) + a_{-1}\left(\frac{G'}{G}\right)^{-1}, \\ \phi_2(\xi) = b_0 + b_1\left(\frac{G'}{G}\right) + b_{-1}\left(\frac{G'}{G}\right)^{-1}. \end{cases} \tag{13}$$

Next, we take the first and second derivatives of ϕ_1 and ϕ_2 using (13), together with (6):

$$\phi_1' = a_1(\alpha-1)\left(\frac{G'}{G}\right)^2 + a_1\beta\left(\frac{G'}{G}\right) + a_1\gamma - a_{-1}(\alpha-1) - a_{-1}\beta\left(\frac{G'}{G}\right)^{-1} - a_{-1}\gamma\left(\frac{G'}{G}\right)^{-2} \tag{14}$$

and

$$\begin{aligned} \phi_1'' = {} & 2a_1(\alpha-1)^2\left(\frac{G'}{G}\right)^3 + 3a_1(\alpha-1)\beta\left(\frac{G'}{G}\right)^2 + \left[2(\alpha-1)\gamma+\beta^2\right]a_1\left(\frac{G'}{G}\right) + \left[a_1\beta\gamma + a_{-1}(\alpha-1)\beta\right] \\ & + a_{-1}\left[\beta^2 + 2(\alpha-1)\gamma\right]\left(\frac{G'}{G}\right)^{-1} + 3a_{-1}\beta\gamma\left(\frac{G'}{G}\right)^{-2} + 2a_{-1}\gamma^2\left(\frac{G'}{G}\right)^{-3}. \end{aligned} \tag{15}$$

Similarly, we can get

$$\phi_2' = b_1(\alpha-1)\left(\frac{G'}{G}\right)^2 + b_1\beta\left(\frac{G'}{G}\right) + b_1\gamma - b_{-1}(\alpha-1) - b_{-1}\beta\left(\frac{G'}{G}\right)^{-1} - b_{-1}\gamma\left(\frac{G'}{G}\right)^{-2} \tag{16}$$

and

$$\begin{aligned} \phi_2'' = {} & 2b_1(\alpha-1)^2\left(\frac{G'}{G}\right)^3 + 3b_1(\alpha-1)\beta\left(\frac{G'}{G}\right)^2 + \left[2(\alpha-1)\gamma+\beta^2\right]b_1\left(\frac{G'}{G}\right) + \left[b_1\beta\gamma + b_{-1}(\alpha-1)\beta\right] \\ & + b_{-1}\left[\beta^2 + 2(\alpha-1)\gamma\right]\left(\frac{G'}{G}\right)^{-1} + 3b_{-1}\beta\gamma\left(\frac{G'}{G}\right)^{-2} + 2b_{-1}\gamma^2\left(\frac{G'}{G}\right)^{-3}. \end{aligned} \tag{17}$$

We then put (14)-(17) into (4), and equating corresponding coefficients of each $\left(\frac{G'}{G}\right)^i$ $(i = 0, \pm1, \pm2, \pm3)$ from (4), we get the polynomial equations:

$$\begin{cases} a_1 = 0 \text{ or } a_1 = \pm\sqrt{2}|\alpha-1| \quad \text{and } a_{-1} = 0 \text{ or } \quad a_{-1} = \pm\sqrt{2}|\gamma|, \\ 3a_1(\alpha-1)\beta + a_1(\alpha-1) + a_1^2\left(r + \frac{1}{\sqrt{\delta}}\right) - a_1b_1 - 3a_0a_1^2 = 0, \\ \left[2(\alpha-1)\gamma+\beta^2\right]a_1 + a_1c\beta - a_1s + 2a_0a_1\left(r + \frac{1}{\sqrt{\delta}}\right) - (a_1b_0 + a_0b_1) - 3a_1(a_0^2 + a_{-1}a_1) = 0, \\ a_1\beta\gamma + a_{-1}(\alpha-1)\beta + a_1\gamma c - a_{-1}(\alpha-1)c - a_0s + (a_0^2 + 2a_{-1}a_1)\left(r + \frac{1}{\sqrt{\delta}}\right) \\ \quad -(a_0b_0 + a_{-1}b_1 + a_1b_{-1}) - a_0(a_0^2 + 6a_{-1}a_1) = 0, \\ a_{-1}\left[\beta^2 + 2(\alpha-1)\gamma\right] - a_{-1}\beta c - a_{-1}s + 2a_{-1}a_0\left(r + \frac{1}{\sqrt{\delta}}\right) - (a_{-1}b_0 + a_0b_{-1}) - 3a_{-1}(a_{-1}a_1 + a_0^2) = 0, \\ 3a_{-1}\beta\gamma - a_{-1}\gamma c + a_{-1}^2\left(r + \frac{1}{\sqrt{\delta}}\right) - a_{-1}b_{-1} - 3a_0a_{-1}^2 = 0 \end{cases} \tag{18}$$

and

$$\begin{cases} b_1 = 0 \text{ or } b_1 = \pm\sqrt{\frac{2}{\delta}}|\alpha-1| \quad \text{and } b_{-1} = 0 \text{ or } \quad b_{-1} = \pm\sqrt{\frac{2}{\delta}}|\gamma|, \\ 3b_1(\alpha-1)\beta + b_1c(\alpha-1) + a_1b_1r - 3b_0b_1^2\delta = 0, \\ \left[2(\alpha-1)\gamma+\beta^2\right]b_1 + b_1\beta c - b_1s + (a_1b_0 + a_0b_1)r - 3b_1(b_0^2 + b_{-1}b_1)\delta = 0, \\ b_1\beta\gamma + b_{-1}(\alpha-1)\beta + \left[b_1\gamma - b_{-1}(\alpha-1)\right]c - b_0s + (a_0b_0 + a_{-1}b_1 + a_1b_{-1})r - b_0(b_0^2 + 6b_{-1}b_1)\delta = 0, \\ b_{-1}\left[\beta^2 + 2(\alpha-1)\gamma\right] - b_{-1}\beta c - b_{-1}s + (a_{-1}b_0 + a_0b_{-1})r - 3b_{-1}(b_{-1}b_1 + b_0^2)\delta = 0, \\ 3b_{-1}\beta\gamma - b_{-1}\gamma c + a_{-1}b_{-1}r - 3b_0b_{-1}^2\delta = 0. \end{cases} \tag{19}$$

Case 1: If $a_1 = a_0 = 0$, then

$$a_{-1} = \frac{(\alpha-1)(\beta-c) - \gamma(3\beta-c)}{r + \frac{1}{\sqrt{\delta}}}.$$

Moreover, if $b_1 = 0$, we have

$$\begin{cases} b_0 = \pm\sqrt{\frac{\gamma(\alpha-1)(\beta-c)(3\beta-c)-s\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right]}{\delta\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right]}} \\ b_{-1} = \pm\frac{\gamma(3\beta-c)}{\delta^{\frac{1}{2}}\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right]^{\frac{3}{2}}}\cdot\sqrt{\gamma(\alpha-1)(\beta-c)(3\beta-c)-s\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right]} \end{cases}.$$

Thus, the solutions are

$$\begin{cases} \phi_1(\xi) = \frac{(\alpha-1)(\beta-c)-\gamma(3\beta-c)}{r+\frac{1}{\sqrt{\delta}}}\left(\frac{G'}{G}\right)^{-1}, \\ \phi_2(\xi) = \pm\frac{\gamma(3\beta-c)}{\delta^{\frac{1}{2}}\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right]^{\frac{3}{2}}}\cdot\sqrt{\gamma(\alpha-1)(\beta-c)(3\beta-c)-s\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right]}\left(\frac{G'}{G}\right)^{-1} \\ \qquad \pm\sqrt{\frac{\gamma(\alpha-1)(\beta-c)(3\beta-c)-s\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right]}{\delta\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right]}}. \end{cases} \tag{20}$$

Case 2: If $a_{-1} = a_0 = 0$, then

$$a_1 = \frac{(\beta+c)\gamma-(\gamma-1)(3\beta-1)}{r+\frac{1}{\sqrt{\delta}}}.$$

Moreover, if $b_{-1} = 0$, we have

$$\begin{cases} b_0 = \pm\sqrt{\frac{\gamma(\alpha-1)(\beta+c)(3\beta+c)-s\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]}{\delta\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]}} \\ b_1 = \pm\frac{(\alpha-1)(3\beta+c)}{\delta^{\frac{1}{2}}\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]^{\frac{3}{2}}}\cdot\sqrt{\gamma(\alpha-1)(\beta+c)(3\beta+c)-s\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]} \end{cases}.$$

Thus, the solutions are

$$\begin{cases} \phi_1(\xi) = \frac{(\beta+c)\gamma-(\gamma-1)(3\beta-1)}{r+\frac{1}{\sqrt{\delta}}}\left(\frac{G'}{G}\right), \\ \phi_2(\xi) = \pm\sqrt{\frac{\gamma(\alpha-1)(\beta+c)(3\beta+c)-s\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]}{\delta\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]}} \\ \qquad \pm\frac{(\alpha-1)(3\beta+c)}{\delta^{\frac{1}{2}}\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]^{\frac{3}{2}}}\cdot\sqrt{\gamma(\alpha-1)(\beta+c)(3\beta+c)-s\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]}\left(\frac{G'}{G}\right). \end{cases} \tag{21}$$

Case 3: If $a_1 = \sqrt{2}(\alpha-1), b_1 = \sqrt{\frac{2}{\delta}}(\alpha-1)$, then

$$\begin{cases} a_0 = \frac{\sqrt{2}(3\beta+1)+2r}{6}, \\ b_0 = \frac{2r+\sqrt{2}(3\beta+c)}{6\sqrt{\delta}} \end{cases}$$

and

$$\begin{aligned} a_{-1} &= \frac{1}{3\sqrt{2}(\alpha-1)}\left\{\left[\beta^2+2(\alpha-1)\gamma-s+c\beta\right]-\frac{1}{36}\left[(3\beta+1)+\sqrt{2}r\right]\left[3\beta+1-5\sqrt{2}r\right]\right. \\ &\quad \left.-\frac{1}{6\sqrt{2\delta}}\left(3\beta-1+2c+\sqrt{2}r\right)\right\}, \\ b_{-1} &= \frac{1}{18\sqrt{2\delta}(\alpha-1)}\left\{6\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right]+\sqrt{2}r(6\beta+c+1)+4r^2-2\left(3\beta+c+\sqrt{2}r\right)^2\right\}. \end{aligned} \tag{22}$$

Hence, the solutions are

$$\begin{cases} \phi_1(\xi) = \frac{\sqrt{2}(3\beta+1)+2r}{6} + \sqrt{2}(\alpha-1)\left(\frac{G'}{G}\right) \\ \qquad + \frac{1}{3\sqrt{2}(\alpha-1)}\left\{\left[\beta^2+2(\alpha-1)\gamma-s+c\beta\right] - \frac{1}{36}\left[(3\beta+1)+\sqrt{2}r\right]\left[3\beta+1-5\sqrt{2}r\right]\right. \\ \qquad \left. - \frac{1}{6\sqrt{2\delta}}\left(3\beta-1+2c+\sqrt{2}r\right)\right\}\left(\frac{G'}{G}\right)^{-1}, \\ \phi_2(\xi) = \frac{2r+\sqrt{2}(3\beta+c)}{6\sqrt{\delta}} + \sqrt{\frac{2}{\delta}}(\alpha-1)\left(\frac{G'}{G}\right) \\ \qquad + \frac{1}{18\sqrt{2\delta}(\alpha-1)}\left\{6\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right] + \sqrt{2}r(6\beta+c+1) + 4r^2 - 2\left(3\beta+c+\sqrt{2}r\right)^2\right\}\left(\frac{G'}{G}\right)^{-1}. \end{cases} \tag{23}$$

Case 4: If $a_{-1} = \sqrt{2}\gamma, b_{-1} = \sqrt{\frac{2}{\delta}}\gamma$, then

$$\begin{cases} a_0 = \frac{1}{6}\left[2r+\sqrt{2}(3\beta-c)\right] \\ b_0 = \frac{\sqrt{2}(3\beta-c)+2r}{6\sqrt{\delta}} \end{cases}$$

and

$$\begin{aligned} a_1 &= \frac{1}{3\sqrt{2}\gamma}\left\{\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right] + \frac{r}{3}\left[2r+\sqrt{2}(3\beta-c)\right] - \frac{1}{12}\left[2r+\sqrt{2}(3\beta-c)\right]^2\right\}, \\ b_1 &= \frac{1}{18\sqrt{2\delta}\gamma}\left\{6\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right] + r\left[2\sqrt{2}(3\beta-c)+4r-2+2\delta^{-\frac{1}{2}}\right] - \left(3\beta-c+\sqrt{2}r\right)^2\right\}. \end{aligned} \tag{24}$$

Hence, the solutions are

$$\begin{cases} \phi_1(\xi) = \sqrt{2}\gamma\left(\frac{G'}{G}\right)^{-1} + \frac{1}{6}\left[2r+\sqrt{2}(3\beta-c)\right] \\ \qquad + \frac{1}{3\sqrt{2}\gamma}\left\{\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right] + \frac{r}{3}\left[2r+\sqrt{2}(3\beta-c)\right] - \frac{1}{12}\left[2r+\sqrt{2}(3\beta-c)\right]^2\right\}\left(\frac{G'}{G}\right), \\ \phi_2(\xi) = \sqrt{\frac{2}{\delta}}\gamma\left(\frac{G'}{G}\right)^{-1} + \frac{\sqrt{2}(3\beta-c)+2r}{6\sqrt{\delta}} \\ \qquad + \frac{1}{18\sqrt{2\delta}\gamma}\left\{6\left[\beta^2+2(\alpha-1)\gamma-s-\beta c\right] + r\left[2\sqrt{2}(3\beta-c)+4r-2+2\delta^{-\frac{1}{2}}\right] - \left(3\beta-c+\sqrt{2}r\right)^2\right\}\left(\frac{G'}{G}\right). \end{cases} \tag{25}$$

Case 5: If $a_1 = \sqrt{2}(\alpha-1), b_1 = -\sqrt{\frac{2}{\delta}}(\alpha-1)$, then

$$\begin{cases} a_0 = \frac{\sqrt{2}(3\beta+1)+2r+4\delta^{-\frac{1}{2}}}{6}, \\ b_0 = \frac{2r-\sqrt{2}(3\beta+c)}{6\sqrt{\delta}} \end{cases}$$

and

$$\begin{aligned} a_{-1} &= \frac{1}{3\sqrt{2}(\alpha-1)}\left\{\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right] + \frac{1}{3}\left(r+\frac{1}{\sqrt{\delta}}\right)\left[\sqrt{2}(3\beta+1)+2r+4\delta^{-\frac{1}{2}}\right]\right. \\ &\quad \left. - \frac{1}{6\sqrt{\delta}}\left[4(r+\delta^{-\frac{1}{2}}) - \sqrt{2}(c-1)\right] - \frac{1}{12}\left[\sqrt{2}(3\beta+1)+2r+4\delta^{-\frac{1}{2}}\right]^2\right\}, \\ b_{-1} &= \frac{1}{18\sqrt{2\delta}(1-\alpha)}\left\{6\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right] + r\left[\sqrt{2}(6\beta+c+1)+4\delta^{-\frac{1}{2}}\right] - (\sqrt{2}r-3\beta-c)^2\right\}. \end{aligned} \tag{26}$$

Hence, the solutions are

$$\begin{cases} \phi_1(\xi) = \frac{\sqrt{2}(3\beta+1)+2r+4\delta^{-\frac{1}{2}}}{6} + \sqrt{2}(\alpha-1)\left(\frac{G'}{G}\right) \\ \qquad + \frac{1}{3\sqrt{2}(\alpha-1)}\left\{\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right] + \frac{1}{3}\left(r+\frac{1}{\sqrt{\delta}}\right)\left[\sqrt{2}(3\beta+1)+2r+4\delta^{-\frac{1}{2}}\right]\right. \\ \qquad \left. - \frac{1}{6\sqrt{\delta}}\left[4(r+\delta^{-\frac{1}{2}}) - \sqrt{2}(c-1)\right] - \frac{1}{12}\left[\sqrt{2}(3\beta+1)+2r+4\delta^{-\frac{1}{2}}\right]^2\right\}\left(\frac{G'}{G}\right)^{-1} \\ \phi_2(\xi) = \frac{2r-\sqrt{2}(3\beta+c)}{6\sqrt{\delta}} - \sqrt{\frac{2}{\delta}}(\alpha-1)\left(\frac{G'}{G}\right) \\ \qquad + \frac{1}{18\sqrt{2\delta}(1-\alpha)}\left\{6\left[\beta^2+2(\alpha-1)\gamma-s+\beta c\right] + r\left[\sqrt{2}(6\beta+c+1)+4\delta^{-\frac{1}{2}}\right] - (\sqrt{2}r-3\beta-c)^2\right\}\left(\frac{G'}{G}\right)^{-1}. \end{cases} \tag{27}$$

Case 6: If $a_{-1} = -\sqrt{2}\gamma, b_{-1} = \sqrt{\frac{2}{\delta}}\gamma$, then

$$\begin{cases} a_0 = \frac{1}{6}\left[2\left(r + \frac{2}{\sqrt{\delta}}\right) - \sqrt{2}(3\beta - c)\right] \\ b_0 = \frac{\sqrt{2}(3\beta - c) - 2r}{6\sqrt{\delta}} \end{cases}$$

and

$$\begin{aligned} a_1 = &-\frac{1}{3\sqrt{2}\gamma}\left\{\left[\beta^2 + 2(\alpha-1)\gamma - s - \beta c\right] + \frac{1}{6\sqrt{\delta}}\left[(4r\sqrt{\delta}+6)\left(r + \frac{2}{\sqrt{\delta}}\right) + 2r\right.\right. \\ &\left.\left. - \sqrt{2}(3\beta - c)(2r\sqrt{\delta}+4)\right] - \frac{1}{12}\left[2\left(r + \frac{2}{\sqrt{\delta}}\right) - \sqrt{2}(3\beta - c)\right]^2\right\}, \\ b_1 = &\frac{1}{18\sqrt{2\delta}\gamma}\left\{6\left[\beta^2 + 2(\alpha-1)\gamma - s - \beta c\right] + r\left[4r - 2\sqrt{2}(3\beta - c) + 4\delta^{-\frac{1}{2}}\right] - (3\beta - c - \sqrt{2}r)^2\right\}. \end{aligned} \tag{28}$$

Hence, the solutions are

$$\begin{cases} \phi_1(\xi) = \frac{1}{6}\left[2(r + \frac{2}{\sqrt{\delta}}) - \sqrt{2}(3\beta - c)\right] - \sqrt{2}\gamma\left(\frac{G'}{G}\right)^{-1} \\ \quad -\frac{1}{3\sqrt{2}\gamma}\left\{\left[\beta^2 + 2(\alpha-1)\gamma - s - \beta c\right] + \frac{1}{6\sqrt{\delta}}\left[(4r\sqrt{\delta}+6)\left(r + \frac{2}{\sqrt{\delta}}\right) + 2r\right.\right. \\ \quad \left.\left. -\sqrt{2}(3\beta - c)(2r\sqrt{\delta}+4)\right] - \frac{1}{12}\left[2\left(r + \frac{2}{\sqrt{\delta}}\right) - \sqrt{2}(3\beta - c)\right]^2\right\}\left(\frac{G'}{G}\right), \\ \phi_2(\xi) = \frac{\sqrt{2}(3\beta - c) - 2r}{6\sqrt{\delta}} + \sqrt{\frac{2}{\delta}}\gamma\left(\frac{G'}{G}\right)^{-1} \\ \quad +\frac{1}{18\sqrt{2\delta}\gamma}\left\{6\left[\beta^2 + 2(\alpha-1)\gamma - s - \beta c\right] + r\left[4r - 2\sqrt{2}(3\beta - c) + 4\delta^{-\frac{1}{2}}\right] - (3\beta - c - \sqrt{2}r)^2\right\}\left(\frac{G'}{G}\right). \end{cases} \tag{29}$$

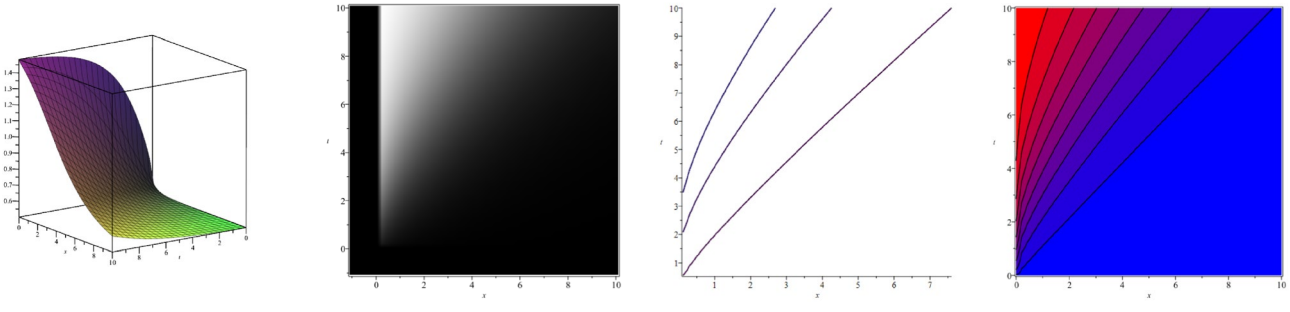
Analysis of figures: solution behavior and fractional derivative variations

In this section, we present a detailed analysis of the solution behaviors for the predator-prey system, as depicted in several figures. The figures provide visual insights into the effects of fractional-order derivatives on the system's dynamics. For simplicity, in all our figures, we may set the parameter values as:

$$b = s = 2, \delta = 0.25.$$

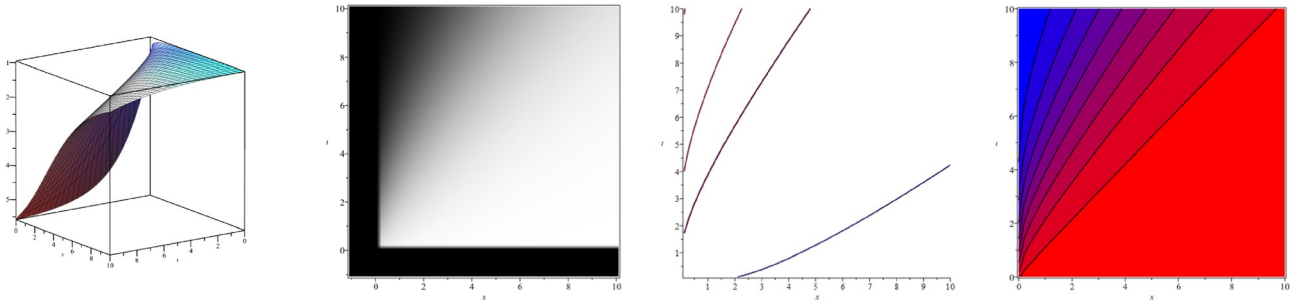
Figures 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18 display 3D, density, and contour plots, where each image corresponds to different parameter values, with the fractional derivative values set at $\mu = \lambda = 0.5$. These plots help us observe the evolution of the predator-prey dynamics, showcasing both the temporal and spatial characteristics of the solutions. In the first two columns of these figures, the 3D and density plots illustrate the changes in the prey and predator densities over time. The final column shows contour maps, where color gradients from blue (low values) to red (high values) represent different density levels, providing an intuitive understanding of how the prey and predator populations evolve under varying conditions.

For instance, in Figures 1a and 2a, the predator population increases, leading to the extinction of the prey population. In contrast, in Figures 7a and 8a, we observe a symbiotic relationship between the predator and prey, where both populations coexist harmoniously. These patterns help us understand how the interaction between the species leads to different equilibrium states, such as extinction or coexistence. Figures 1b-d, 2b-d, 3b-d, 4b-d, 5b-d, 6b-d, 7b-d, 8b-d, 9b-d, 10b-d, 11b-d, 12b-d, 13b-d, 14b-d, 15b-d, 16b-d, 17b-d, 18b-d denote the density



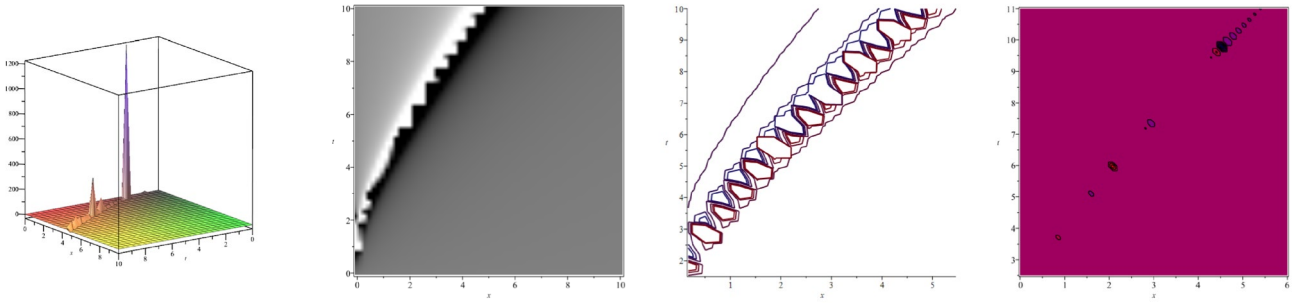
(a) (b) (c) (d)

Fig. 1. $\alpha = 1.5, \beta = 2, \gamma = 1$ $(\Delta > 0), C_1 = 2C_2, \phi_1$ in (20).



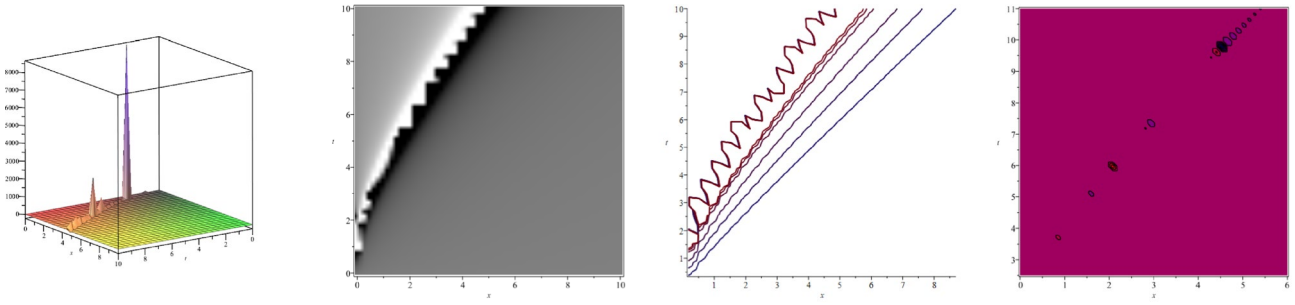
(a) (b) (c) (d)

Fig. 2. $\alpha = 1.5, \beta = 2, \gamma = 1$ $(\Delta > 0), , C_1 = 2C_2, \phi_2$ in (20).



(a) (b) (c) (d)

Fig. 3. $\alpha = \beta = 2, \gamma = 0.5$ $(\Delta = 0), C_1 = C_2, \phi_1$ in (20).



(a) (b) (c) (d)

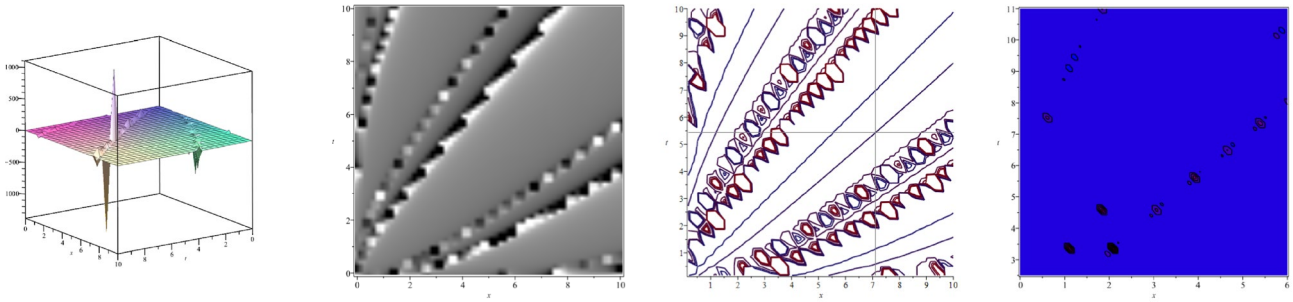
Fig. 4. $\alpha = \beta = 2, \gamma = 0.5$ $(\Delta = 0), C_1 = C_2, \phi_2$ in (20).

map, and the corresponding contour maps, respectively. Figures 3a and 4a, Figures 5a and 6a, up to Figures 17a and 18a denote the 3D evolution pattern between prey and predator.

Furthermore, Figures 19 through 21 illustrate the impact of varying the fractional order derivative on the solutions. As the fractional order approaches 0, we observe that the singularities of the solutions shift towards the boundary of the region. In contrast, as the fractional order approaches 1, the singularities move towards the angle bisector of space-time, resulting in a more stable solution in other regions. Figure 19a-i, 20a-i, 21a-i depict the evolution of 3D images, while Figures 19j-l , 20j-l, 21j-l illustrate the impact on the prey density ϕ_1 when $\mu = 0$ is held constant and λ is varied. These observations underline the crucial role that the fractional derivative plays in shaping the system's behavior.

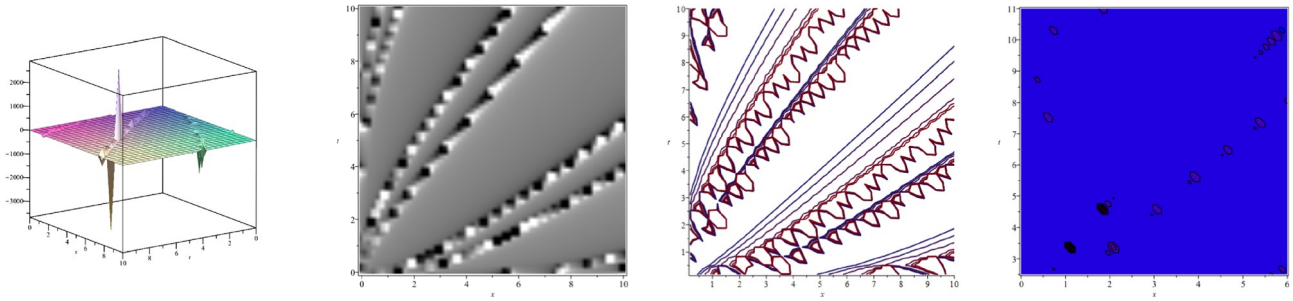
In particular, Figure 21f demonstrates the sensitivity of the solution when the fractional derivative is near 0.7. Small changes in the fractional order at this point lead to significant alterations in the solution, inducing chaotic behavior. This finding is particularly noteworthy, as it highlights a threshold effect where the system's dynamics become highly sensitive to fractional order variations.

Lastly, the images in Figure 22 further illustrate the influence of fractional derivatives on the solution's regularity. As the fractional order transitions near 0.7, the solutions exhibit a distinct shift, from 3D images in



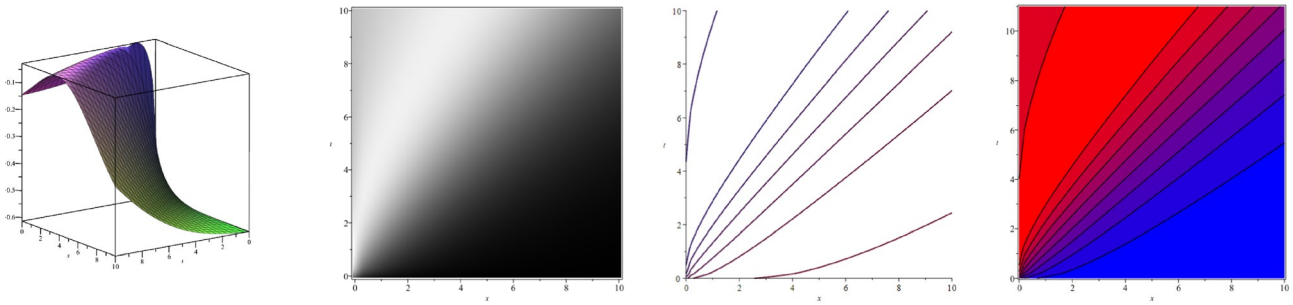
(a) (b) (c) (d)

Fig. 5. $\alpha = \beta = 2\ (\Delta < 0), \gamma = 1, C_2 = 0, \phi_1$ in (20).



(a) (b) (c) (d)

Fig. 6. $\alpha = \beta = 2, \gamma = 1\ (\Delta < 0), C_2 = 0, \phi_2$ in (20).



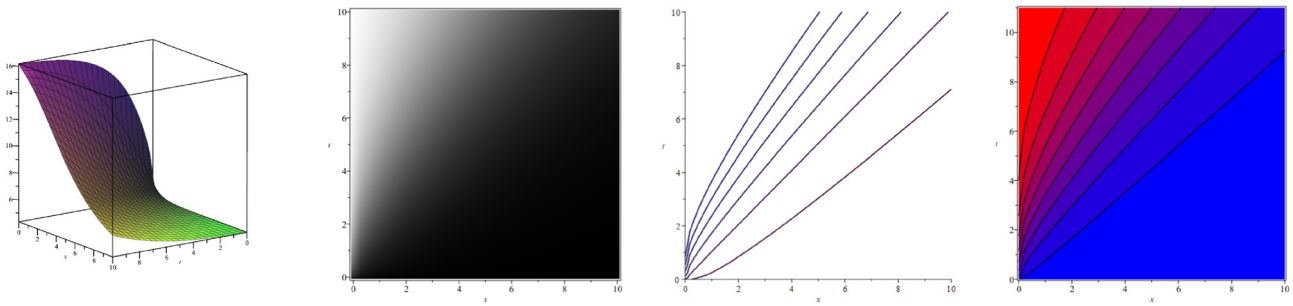
(a) (b) (c) (d)

Fig. 7. $\alpha = 1.5, \beta = 2, \gamma = 1\ (\Delta > 0), C_1 = 2C_2, \phi_1$ in (23).

Figures 22a-h with μ and λ shifting around 0.7, grayscale images in Figure 22i,m,q to contour lines in Figure 22j-l,n-p,r-t, emphasizing the critical impact of fractional derivatives on the system's overall behavior. These graphs visually capture how the predator and prey populations are influenced by the fractional-order dynamics, reinforcing the importance of fractional derivatives in understanding complex ecological systems.

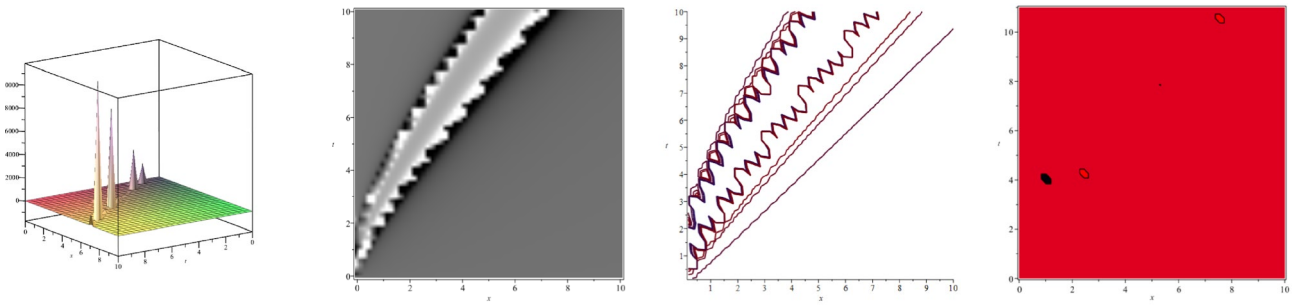
The images (a) through (i) in figure 19-21 show the change in the images corresponding to different fractional derivatives when $\Delta > 0$, $\Delta = 0$, and $\Delta < 0$, respectively.

The grayscale and contour plots in the image depict how a mathematical function or system behaves as parameters μ and λ approach a critical value (likely near 0.7). Grayscale plots (density maps) use color intensity to show function values, where darker/lighter regions indicate lower/higher values. As $\mu = \lambda = 0.7$, the grayscale patterns reveal how small parameter changes affect the system's output, highlighting potential phase transitions or bifurcations. Contour plots overlay lines of equal value (isolines) on the grayscale data, precisely marking thresholds or critical regions. The convergence of contours near $\mu = \lambda = 0.7$ suggests a sensitive dependence on parameters, possibly indicating stability changes or singularities. We make the following key observation: The sequence demonstrates how tiny parameter adjustments (e.g.,$0.701 \rightarrow 0.700001$) can dramatically alter system



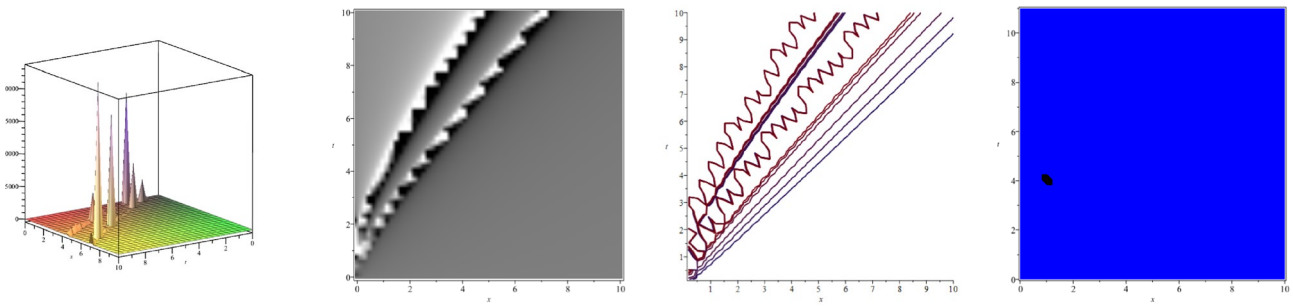
(a) (b) (c) (d)

Fig. 8. $\alpha = 1.5, \beta = 2, \gamma = 1$ $(\Delta > 0), C_1 = 2C_2, \phi_2$ in (23).



(a) (b) (c) (d)

Fig. 9. $\alpha = \beta = 2, \gamma = 0.5$ $(\Delta = 0), C_1 = C_2, \phi_1$ in (23).



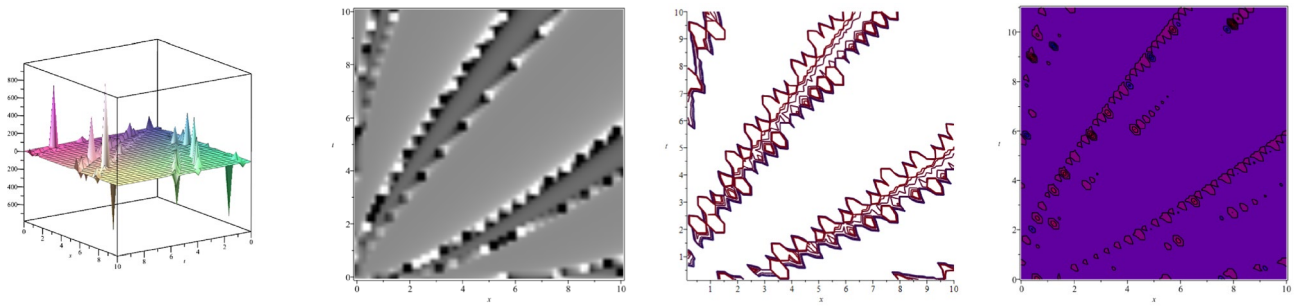
(a) (b) (c) (d)

Fig. 10. $\alpha = \beta = 2, \gamma = 0.5$ $(\Delta = 0), C_1 = C_2, \phi_2$ in (23).

behavior, a hallmark of nonlinear dynamics or critical phenomena. The repeated values for 0.69 and 0.7 may represent control cases for comparison.

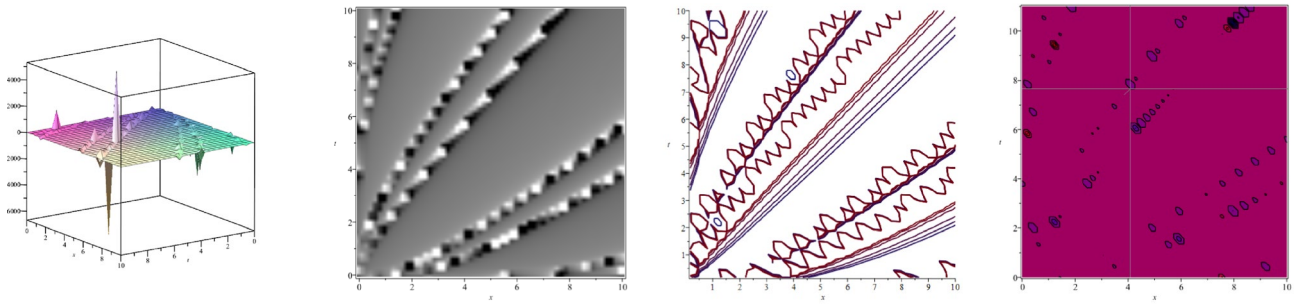
Conclusion

The application of the extended-$\left(\frac{G'}{G}\right)$-expansion method to the conformable space-time fractional diffusive predator-prey system has led to significant advances in understanding the dynamics of biological models. The paper highlights the introduction of negative exponents to generate a broader range of exact solutions, which offers a more complete and precise view of the system's behavior. One key innovation is the discovery of how the solutions' singularities move depending on fractional orders, with a shift towards the boundary when the fractional orders decrease (i.e. $\mu, \lambda \to 0$) and towards the angle bisector of space-time as they approach 1 (i.e. $\mu, \lambda \to 1$). Furthermore, the analysis of chaotic phenomena occurring around a fractional order $\mu = \lambda = 0.7$ provides intriguing insights into the sensitivity of the system, where small changes in the fractional derivative result in dramatic shifts in the solution. These findings not only enrich the mathematical modeling of predator-



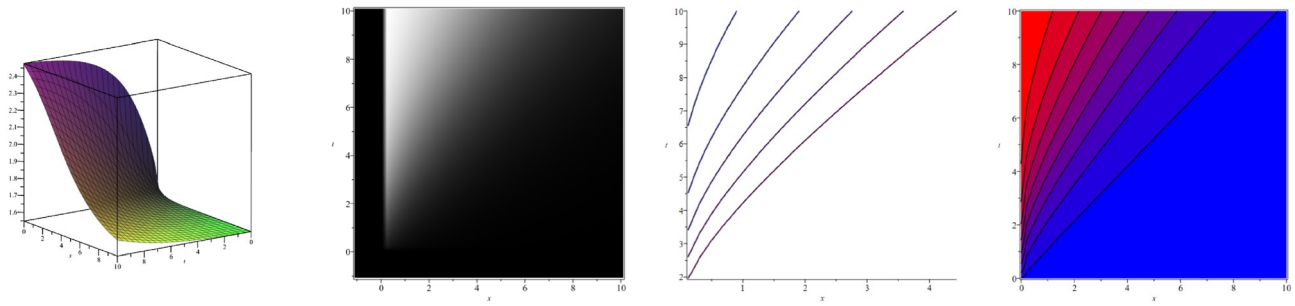
(a) (b) (c) (d)

Fig. 11. $\alpha = \beta = 2, \gamma = 1\ (\Delta < 0), C_2 = 0, \phi_1$ in (23).



(a) (b) (c) (d)

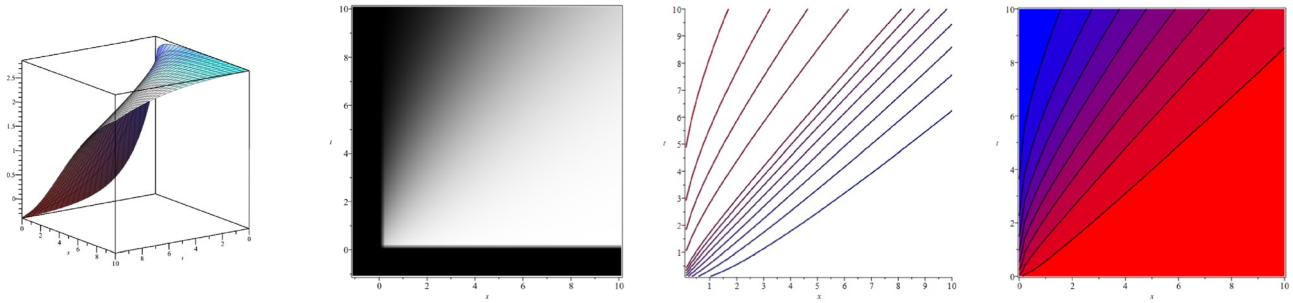
Fig. 12. $\alpha = \beta = 2, \gamma = 1\ (\Delta < 0), C_2 = 0, \phi_2$ in (23).



(a) (b) (c) (d)

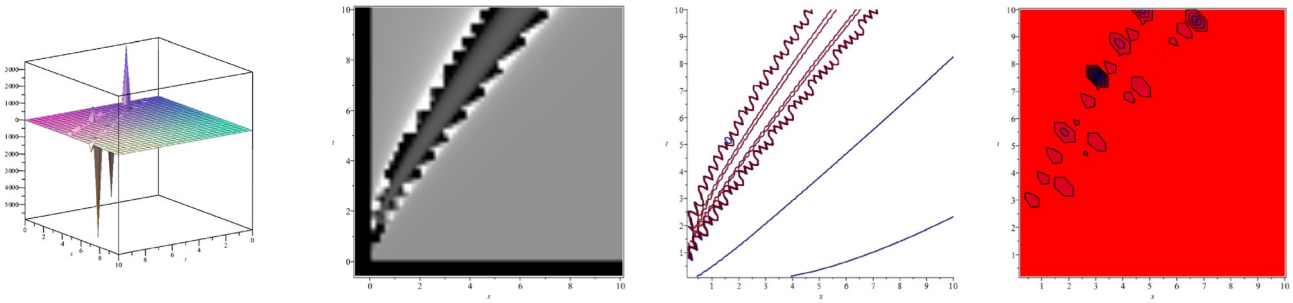
Fig. 13. $\alpha = 1.5, \beta = 2, \gamma = 1\ (\Delta > 0), C_1 = 2C_2, \phi_1$ in (29).

prey systems but also offer valuable implications for ecological studies and further exploration of fractional differential equations in dynamic systems. The essence of the $(\frac{G'}{G})$ method expansion is to use the highest order derivative term to balance the highest order nonlinear term, thereby forming a closed loop, which explains why convergence and stability could be established. For future work, we may consider the chaos phenomenon around $\mu = \lambda = 0.7$, and here we can test the chaos by calculating the Lyapunov exponent around $\mu = \lambda = 0.7$. Through observation of the figures, we can see that the regularity of the solutions near the folding region is poor. In the future, we can consider using a prior estimate to obtain useful energy functionals, and using Sobolev space theory to obtain some mathematically regular solutions. The solutions to these problems are helpful for a better understanding of biology and physics.



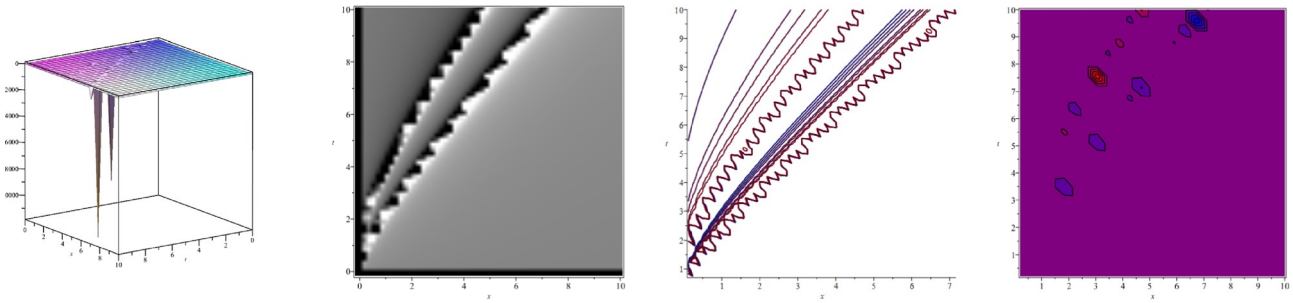
(a) (b) (c) (d)

Fig. 14. $\alpha = 1.5, \beta = 2, \gamma = 1$ $(\Delta > 0), C_1 = 2C_2, \phi_2$ in (29).



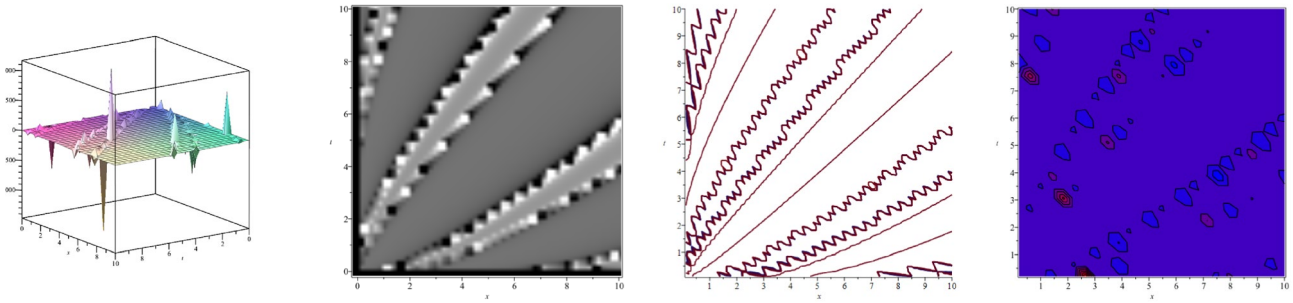
(a) (b) (c) (d)

Fig. 15. $\alpha = \beta = 2, \gamma = 0.5$ $(\Delta = 0), C_1 = C_2, \phi_1$ in (29).



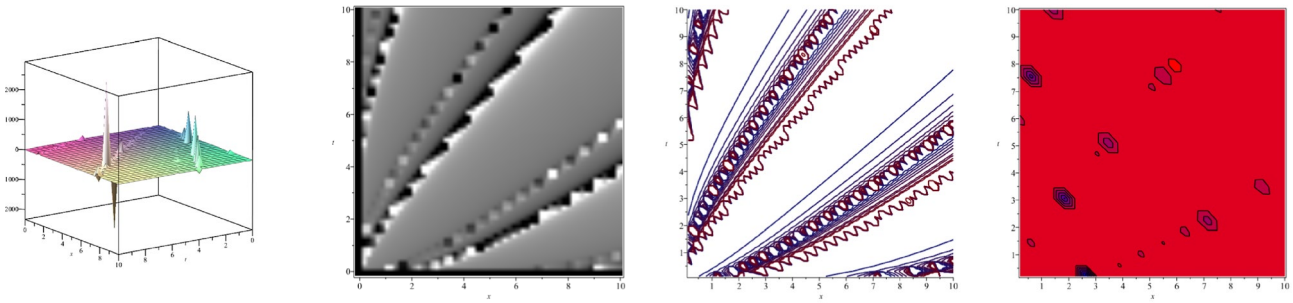
(a) (b) (c) (d)

Fig. 16. $\alpha = \beta = 2, \gamma = 0.5$ $(\Delta = 0), C_1 = C_2, \phi_2$ in (29).



(a) (b) (c) (d)

Fig. 17. $\alpha = \beta = 2, \gamma = 1$ $(\Delta < 0), C_2 = 0, \phi_1$ in (29).



(a) (b) (c) (d)

Fig. 18. $\alpha = \beta = 2, \gamma = 1$ $(\Delta < 0), C_2 = 0, \phi_2$ in (29).

$\Delta = \beta^2 - 4\alpha(\alpha - 1)\gamma$	The values of parameters	The solutions of ϕ_i $(i = 1, 2)$	Graph
$\Delta > 0$	$\alpha = 1.5, \beta = 2, \gamma = 1, C_1 = 2C_2$	ϕ_i in (20)	Figure 1-2
		ϕ_i in (23)	Figure 7-8
		ϕ_i in (29)	Figure 13-14
$\Delta = 0$	$\alpha = \beta = 2, \gamma = 0.5, C_1 = C_2$	ϕ_i in (20)	Figure 3-4
		ϕ_i in (23)	Figure 9-10
		ϕ_i in (29)	Figure 15-16
$\Delta < 0$	$\alpha = \beta = 2, \gamma = 1, C_2 = 0$	ϕ_i in (20)	Figure 5-6
		ϕ_i in (23)	Figure 11-12
		ϕ_i in (29)	Figure 17-18

Table 1. Figure 1–18 represent the graphs of taking the different solutions 20, 23, and 29 under the $\Delta > 0, \Delta = 0$, and $\Delta < 0$.

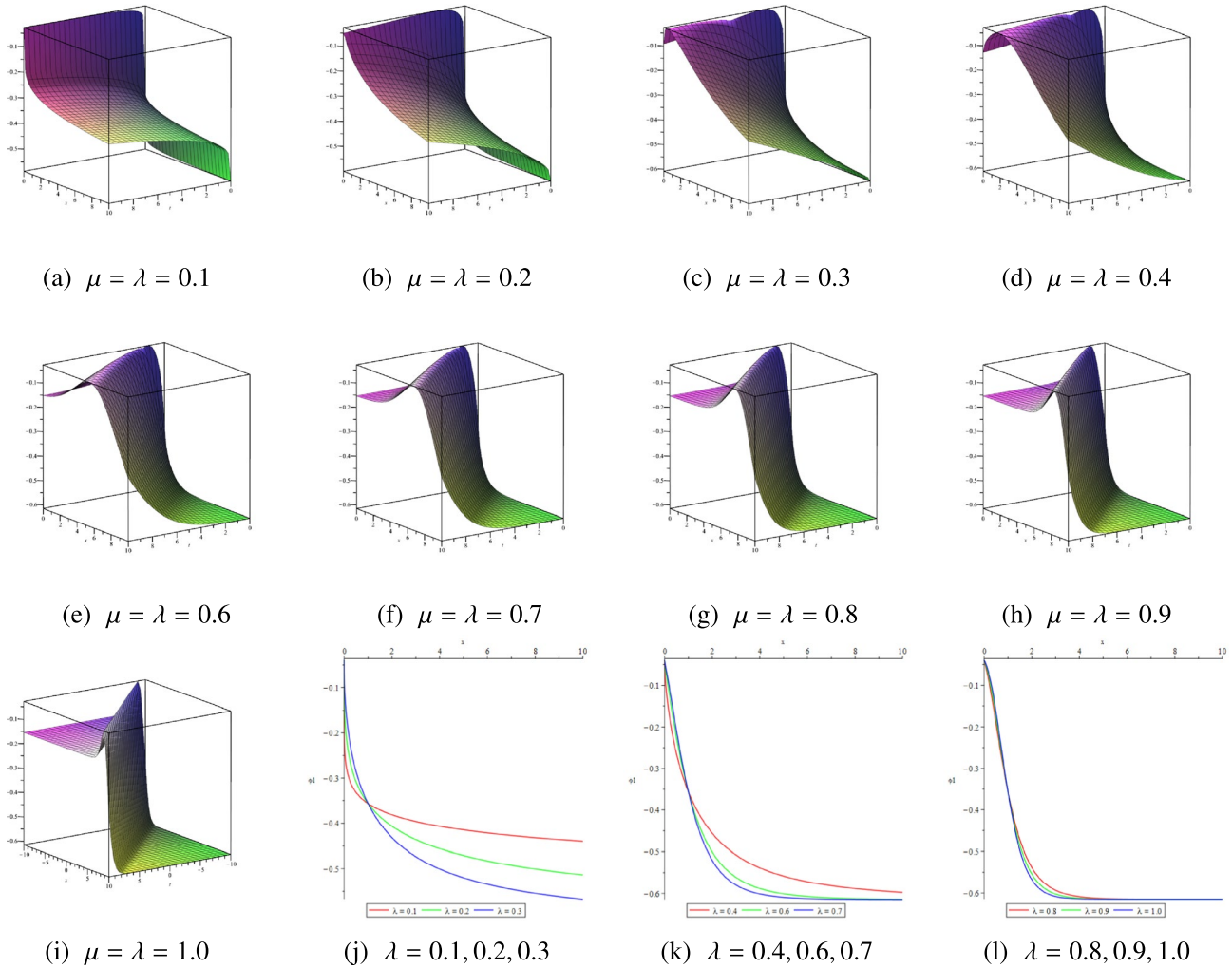
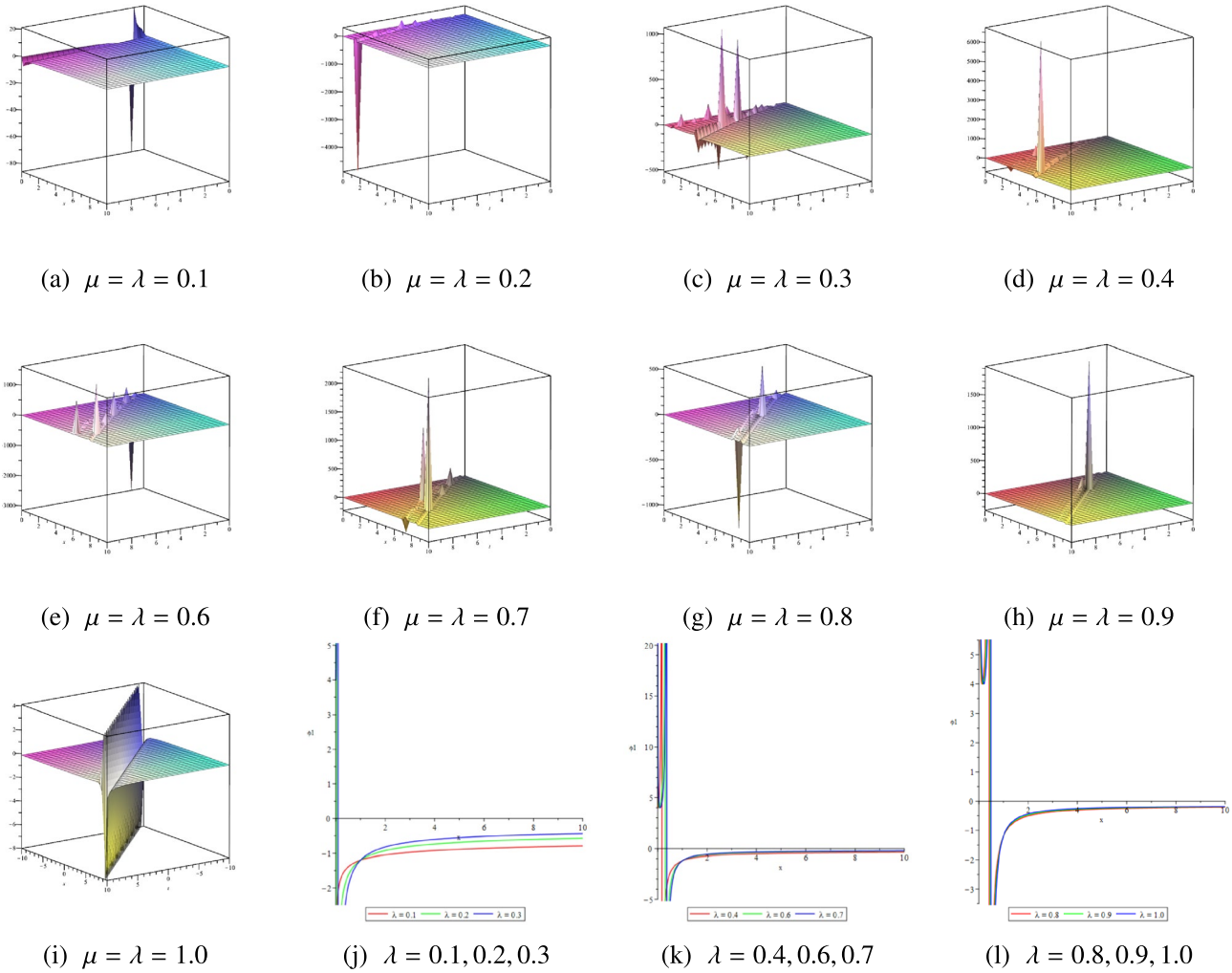


Fig. 19. (**a**)-(**i**) The 3D graphs of $\phi_1(x,t)$ for distinct values of $\mu = \lambda$ and $\Delta > 0$ in (23), (**j**)-(**l**) The 2D graphs of $\phi_1(x)$ for different $\lambda, t \equiv 1$and $\Delta > 0$ in (23).

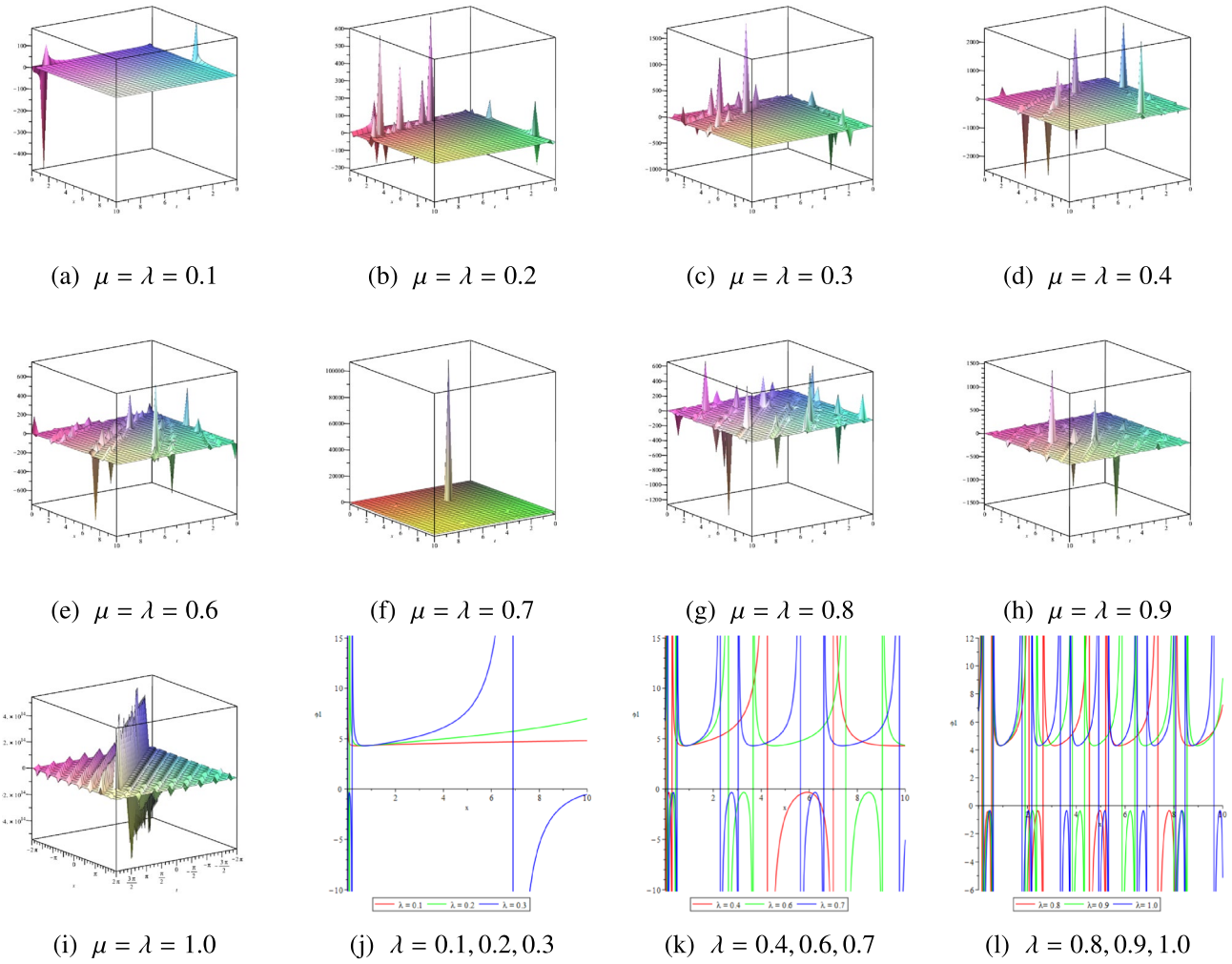


(a) $\mu = \lambda = 0.1$ (b) $\mu = \lambda = 0.2$ (c) $\mu = \lambda = 0.3$ (d) $\mu = \lambda = 0.4$

(e) $\mu = \lambda = 0.6$ (f) $\mu = \lambda = 0.7$ (g) $\mu = \lambda = 0.8$ (h) $\mu = \lambda = 0.9$

(i) $\mu = \lambda = 1.0$ (j) $\lambda = 0.1, 0.2, 0.3$ (k) $\lambda = 0.4, 0.6, 0.7$ (l) $\lambda = 0.8, 0.9, 1.0$

Fig. 20. (**a**)-(**i**) The 3D graphs of $\phi_1(x,t)$ for distinct values of $\mu = \lambda$ and $\Delta = 0$ in (23), (**j**)-(**l**) The 2D graphs of $\phi_1(x)$ for different $\lambda, t \equiv 1$and $\Delta = 0$ in (23).

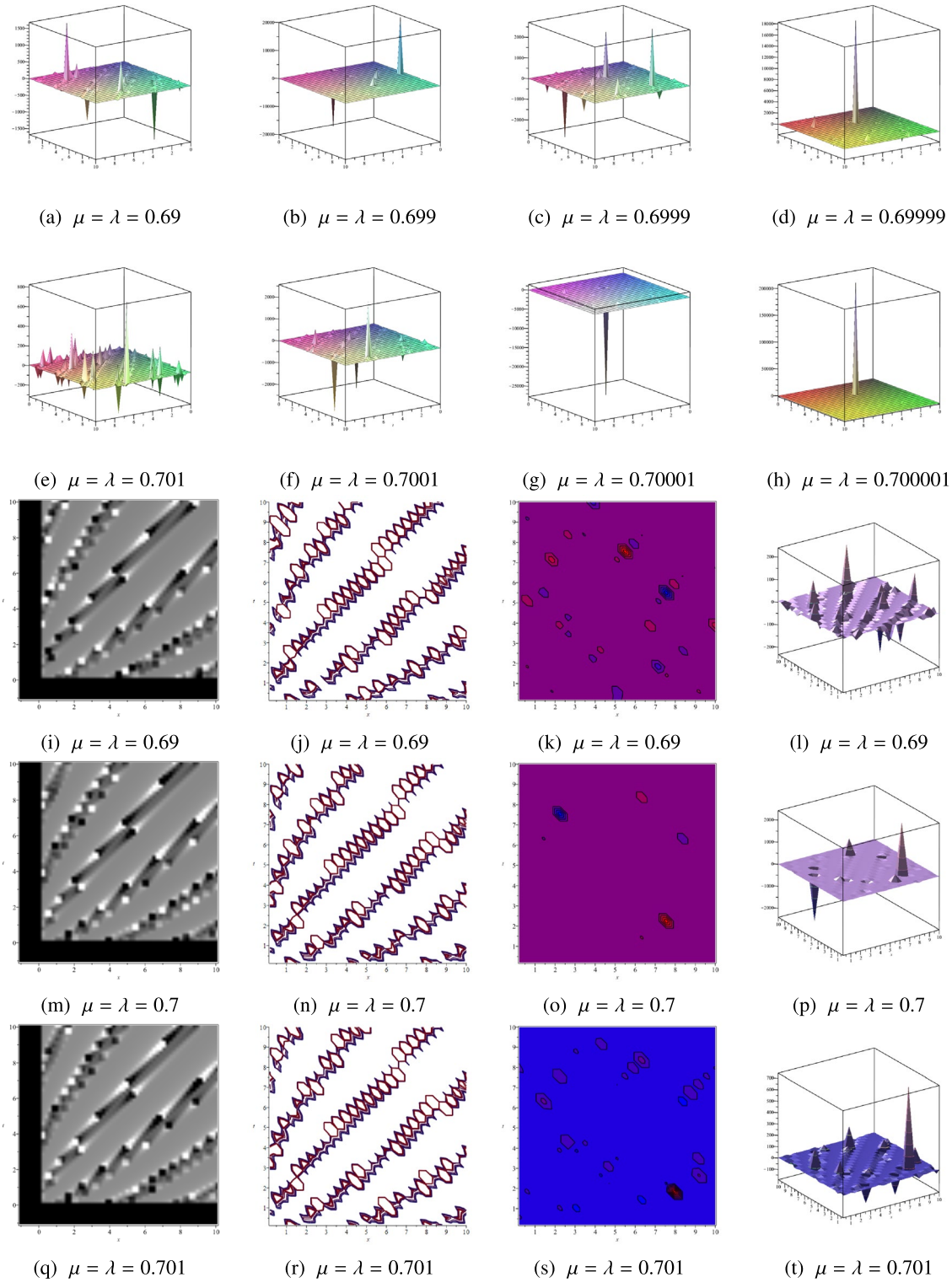


(a) $\mu = \lambda = 0.1$ (b) $\mu = \lambda = 0.2$ (c) $\mu = \lambda = 0.3$ (d) $\mu = \lambda = 0.4$

(e) $\mu = \lambda = 0.6$ (f) $\mu = \lambda = 0.7$ (g) $\mu = \lambda = 0.8$ (h) $\mu = \lambda = 0.9$

(i) $\mu = \lambda = 1.0$ (j) $\lambda = 0.1, 0.2, 0.3$ (k) $\lambda = 0.4, 0.6, 0.7$ (l) $\lambda = 0.8, 0.9, 1.0$

Fig. 21. (**a**)-(**i**) The 3D graphs of $\phi_1(x,t)$ for distinct values of $\mu = \lambda$ and $\Delta < 0$ in (23), (**j**)-(**l**) The 2D graphs of $\phi_1(x)$ for different $\lambda, t \equiv 1$and $\Delta < 0$ in (23).



(a) $\mu = \lambda = 0.69$ (b) $\mu = \lambda = 0.699$ (c) $\mu = \lambda = 0.6999$ (d) $\mu = \lambda = 0.69999$

(e) $\mu = \lambda = 0.701$ (f) $\mu = \lambda = 0.7001$ (g) $\mu = \lambda = 0.70001$ (h) $\mu = \lambda = 0.700001$

(i) $\mu = \lambda = 0.69$ (j) $\mu = \lambda = 0.69$ (k) $\mu = \lambda = 0.69$ (l) $\mu = \lambda = 0.69$

(m) $\mu = \lambda = 0.7$ (n) $\mu = \lambda = 0.7$ (o) $\mu = \lambda = 0.7$ (p) $\mu = \lambda = 0.7$

(q) $\mu = \lambda = 0.701$ (r) $\mu = \lambda = 0.701$ (s) $\mu = \lambda = 0.701$ (t) $\mu = \lambda = 0.701$

Fig. 22. (**a**)-(**h**) The graphs of ϕ_1 in (23) for various values of $\mu = \lambda$ near 0.7, (**i**) (**m**) (**q**) The density of ϕ_1 in (23) for $\mu = 0.69, 0.7, 0.701$, (**j**)-(**l**), (**n**)-(**p**), (**r**)-(**t**) The contours of ϕ_1 in (23) for $\mu = 0.69, 0.7, 0.701$.

Data availability

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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Author contributions

J. W. and Z. Y. were responsible for the writing, computation, and analytical reasoning. Z. L. conceptualized the ideas for this article, while H. T. contributed to the discussion and critical review of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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